

Auto Ticket Classification using Flow Designer

INTRODUCTION:

Team ID : SWTID-2026-3996

Team Size : 4

Team Leader : Dharsan A N S

Team member : Aravindh Kumar M

Team member : Dinesh Kumar T

Team member : Manojpandi S

Problem Statement

The school IT helpdesk receives multiple incident requests daily from students and teachers, such as Wi-Fi issues, projector failures, password problems, and slow computers. Currently, IT staff manually reviews each request and assigns a category, which is time-consuming and inefficient.

This project automates ticket classification using ServiceNow Flow Designer by analyzing keywords in the incident's short description and description fields.

Objective

- Automatically classify incidents at the time of creation.
- Reduce manual effort for IT agents.
- Improve ticket routing efficiency.
- Implement a no-code, easily maintainable solution.

PHASE 1: Requirement Analysis & Planning

The school IT helpdesk receives multiple support requests daily from students and teachers related to common issues such as network connectivity, hardware failures, account access problems, and system performance issues.

Currently, IT staff manually review each ticket to:

- Identify the issue type.
- Select the correct category and subcategory.
- Inform the caller about ticket creation.

This manual process is time-consuming, error-prone, and not scalable as the number of requests increases.

Activity : 1 Business Use Case.

Business Requirements

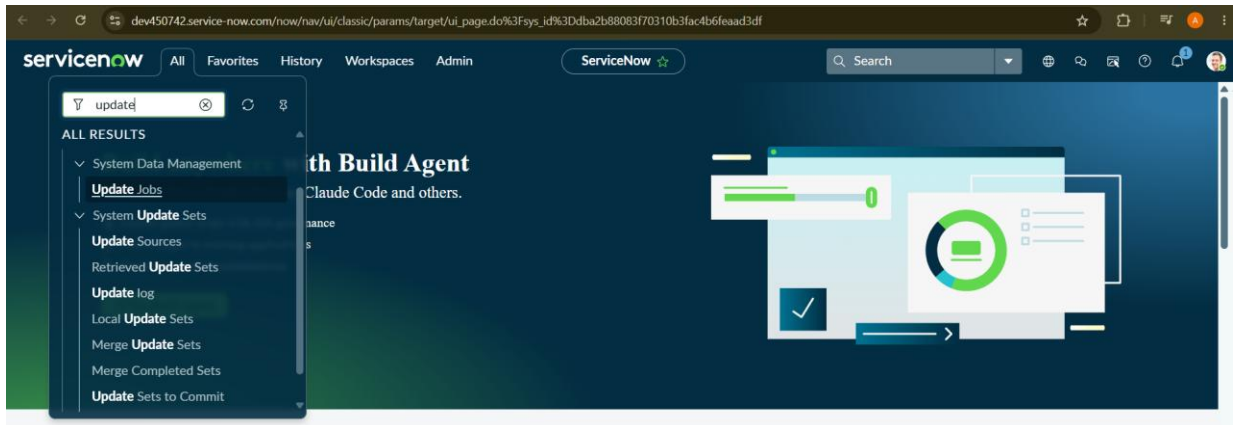
The system must:

1. Automatically classify IT tickets based on the issue description.
2. Assign both Category and Subcategory without manual intervention.
3. Support dependent choice logic between Category and Subcategory.
4. Send an automated email notification to the caller upon ticket creation.
5. Store ticket information in a structured and standardized format.
6. Ensure easy maintenance and future scalability.

Activity : 2 Creation of New Update Set.

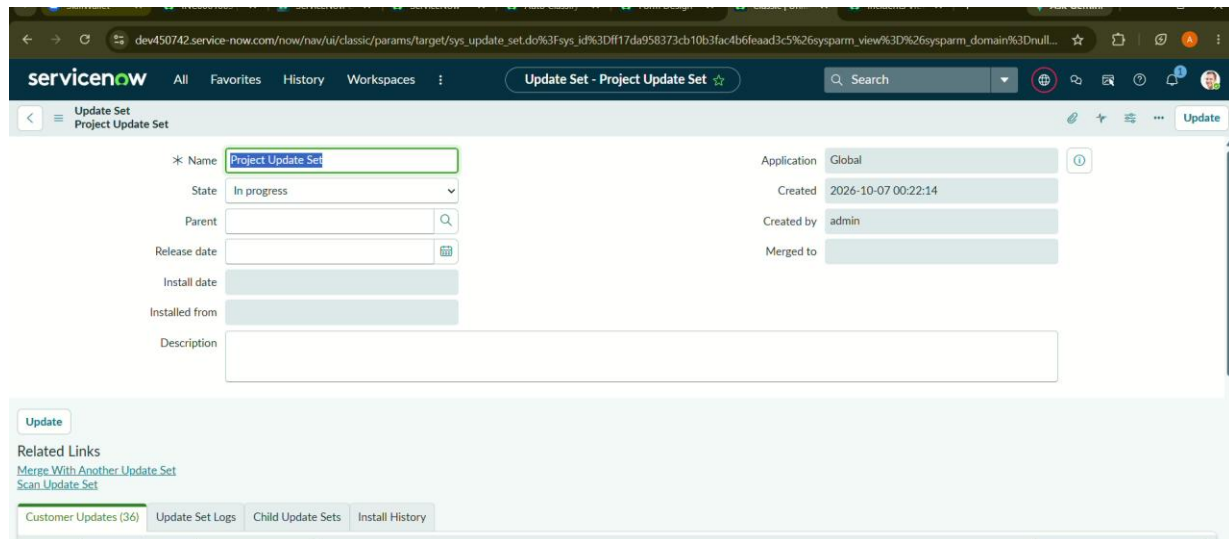
Navigation Steps :

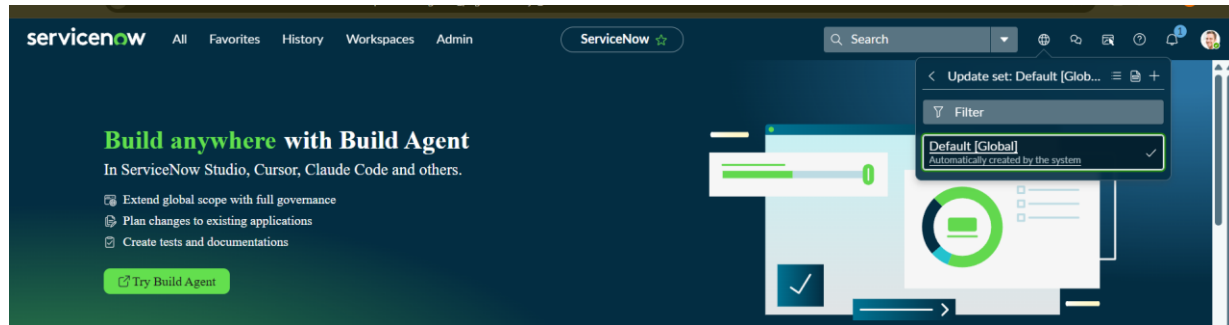
- Go to All menus → Search for Update set.
- Under System Update Sets → Select Local Update Sets.
- Click the New button on the top right corner of the page.
- Name : **Project Update Set.**
- State : **In progress.**
- Application : **Global.** ● Then **Save the Form.**



- Under Related links Click on the **Make This My Current Set**. • Then the Created Update Set will appear as Current.

Activity : 3 Navigation steps in Picture View.



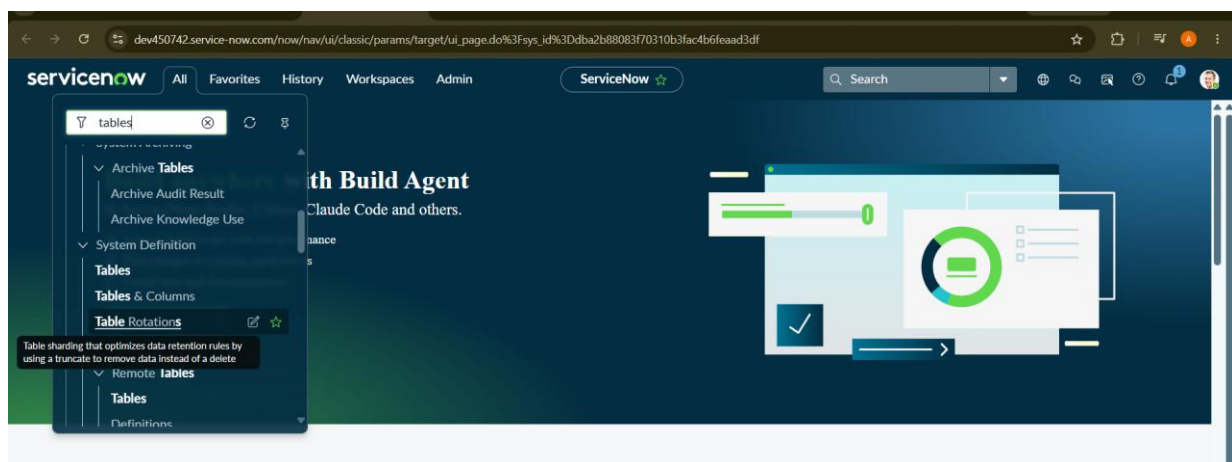


Phase :- 2 Backend Development & Configuration

Activity 1 : Creation of Custom Table to Store Ticket Records.

Steps to Create Custom Table

- Open All menus → Search for Tables.
- Under **System Definition** → Click **Tables**.
- On the top right corner Click **New** button.
- Label : **Incident WorkFlow**.
- Uncheck the **Create module**.
- Now **Save** the Form.
- Then Click the **Controls** Suction and Enable the Check box **true Auto Number**.
- Now again **save** the Form.



dev450742.service-now.com/now/nav/ui/classic/params/target/sys_db_object_list.do%3Fsysparm_first_row%3D1%26sysparm_query%3Dsys_update_name!SNOTEMPTY%25SeGOTOlabel...

servicenow All Favorites History Workspaces Admin Tables Search

Label Search

Actions on selected rows... New

All > Update name is not empty > Label > Incident workflow

Label	Name	Extends table	Extensible	Updated
Search	Search	Search	Search	Search
Incident WorkFlow	u_incident_workflow	(empty)	false	2026-10-07 02:08:25
Inclusion Endpoint	cmdb_ci_endpoint_inclusion	Endpoint	true	2026-06-12 14:06:28
Incoming App Versions	sn_um_incoming_app_version	(empty)	false	2026-06-12 15:08:39
Incoming Store Apps	sn_um_incoming_app	(empty)	false	2026-06-12 15:08:37
Incompatible Guarded Scripts	sys_gs_incompatible_scripts	(empty)	false	2026-06-12 13:57:31
Incremental Offline Deny List	sys_sg_incremental_offline_exclusion_list	(empty)	false	2026-06-12 15:01:24
Incremental offline result	sys_sg_incremental_result	(empty)	false	2026-06-12 15:01:24
Index Explanation	sys_index_explain	MySQL Explain Plan Step	false	2026-06-12 14:04:05
Index Hint Rewrite	sys_query_index_hint	Query Rewrite	false	2026-06-12 14:19:48
Index History	sn_aisearch_global_job_ingest_history	(empty)	false	2026-06-12 15:02:19
Index Suggestion	sys_index_suggestion	(empty)	false	2026-06-12 14:04:05
Index Suggestion Executions	sys_index_suggestion_exec	(empty)	false	2026-06-12 14:04:05

dev450742.service-now.com/now/nav/ui/classic/params/target/sys_db_object_list.do%3Fsysparm_first_row%3D1%26sysparm_query%3Dsys_update_name!SNOTEMPTY%25SeGOTOlabel...

servicenow All Favorites History Workspaces Admin Tables Search

Table Incident WorkFlow Delete Update Delete All Records

* Label Incident WorkFlow Application Global

* Name u_incident_workflow Remote Table

Columns Controls Application Access

Table Columns for text Search 1 to 15 of 15 New

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Subcategory	String	(empty)	40		false
Short Description	String	(empty)	40		false
Updated by	String	(empty)	40		false
Created by	String	(empty)	40		false
State	String	(empty)	40		false

Incident WorkFlow Update Delete

Save

Insert

Insert and Stay

Analyze Access

Configure

Export

View

Create Favorite

Copy URL

Copy sys_id

Show XML

History

Reload form

Form Builder

Form Design

Form Layout

Related Lists

All

Table

Security Rules

Business Rules

Client Scripts

UI Policies

Data Policies

UI Actions

Notifications

Dictionary

Number INC00519

* Caller Janice Twiet

State -- None --

Assigned group

Assigned to

Update Delete

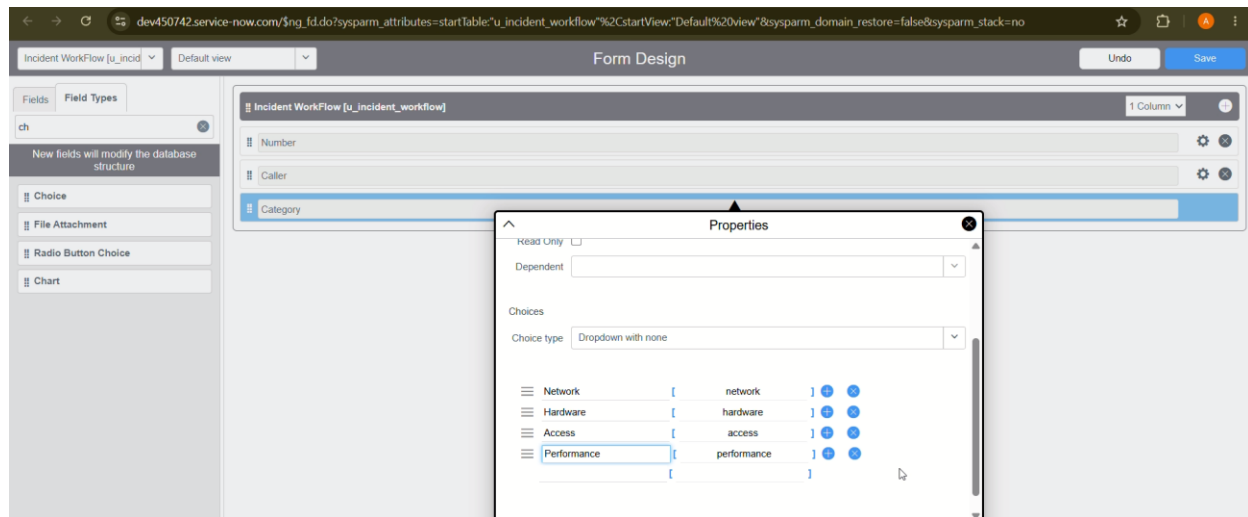
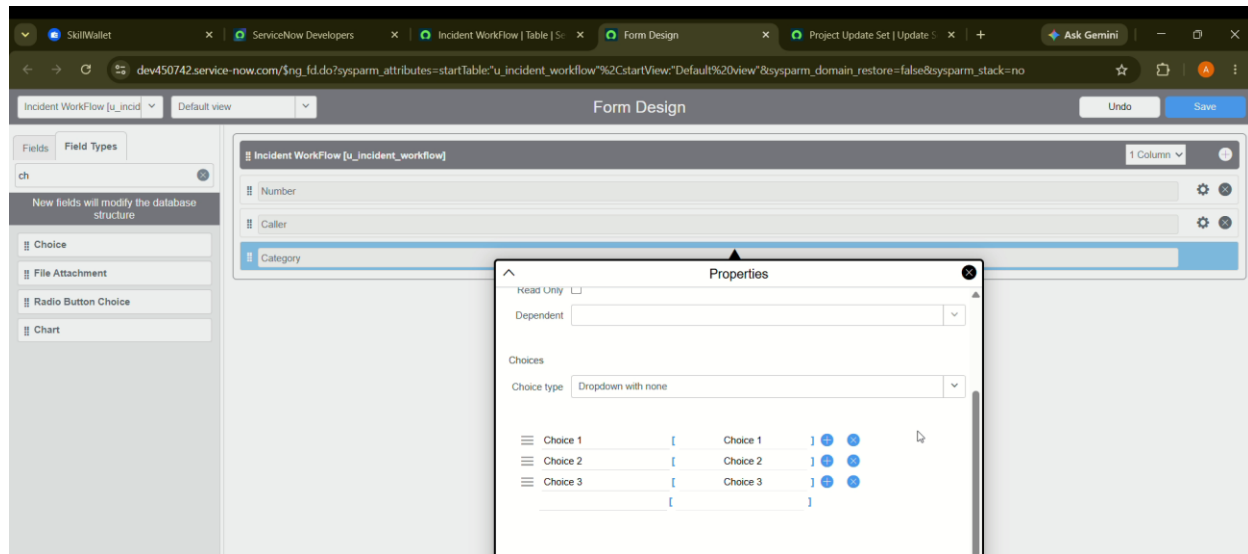
-
- Then Save the Form and Under Related Links Click **Show Form**.
 - Now Open the Configure On clicking the Context Menu and then Open the **Form Designer**.
 - Then Create Fields as Required.

Activity 2: Field Creation and Data Type Configuration

The objective of this milestone is to create all required fields in the School IT Ticket table with correct data types, choices, and dependencies, ensuring structured and consistent data capture.

Field Label	Type	Reference/Choice
Number	Auto Number	—
Caller	Reference	sys_user
Category	Choice	[Network,Hardware,Access,Performance]
Subcategory	Choice	[Wi-Fi,Projector,Forgot Password,Slow Computer]
Short Description	String	—
Description	String	—
State	Choice	[New,In progress,On hold,Resolved,Closed]
Assigned Group	Reference	sys_user_group
Assigned to	Reference	sys_user

- After Creating Fields with their appropriate Type, then add the Choice from the Form design.
- To all the Choice list Fields.
- They are Category,Subcategory and State.



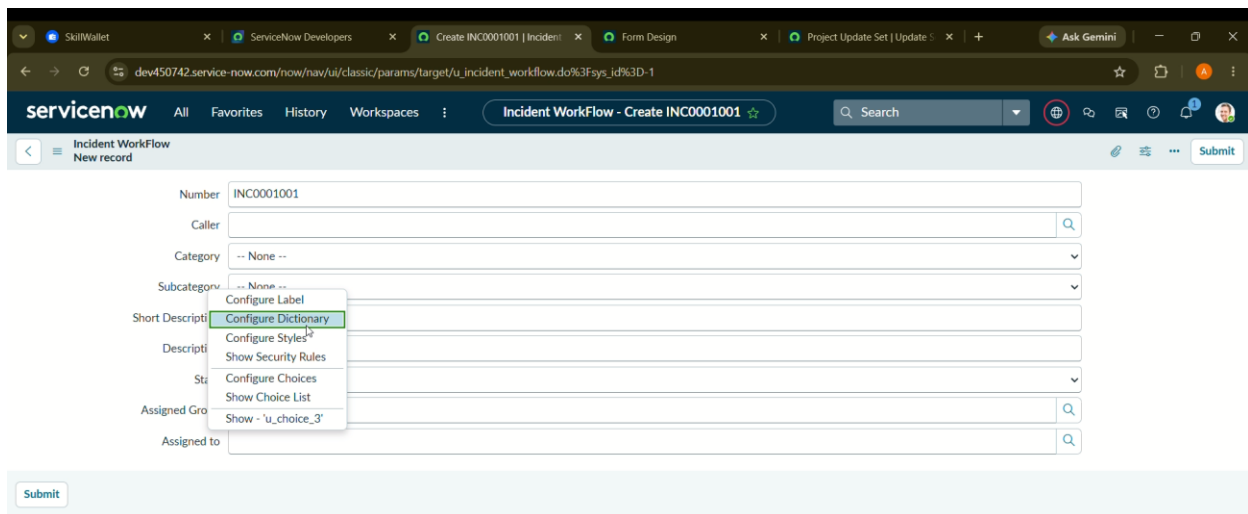
- Same like that, add Choice to the State field.
- Then Click the save button on the top right corner.
- Open the Form and reload it.

Activity 3 : Implementing Dependency Between Category and Subcategory Choice Fields

The objective of this milestone is to implement a **dependency relationship** between the **Category** and **Subcategory** fields so that the available Subcategory options dynamically change based on the selected Category.

Navigation Steps :

- Open the Created Table Form and Right click on the **Subcategory** Field.
- Click on the **Configure Dictionary**.
- Under **Related links** Click on the **Advanced view**, ● Then select the **Dependent Field** Section.
- Enable the checkbox **Use dependent field true**.
- Then select the Field **Category** on the **Dependent on field**.
- Then **save** the Form.



The screenshot shows the ServiceNow web interface for an Incident WorkFlow form titled 'Create INC0001001'. The form includes fields for Number, Caller, Category, Subcategory, Short Description, Description, State, Assigned Group, and Assigned to. A context menu is open over the Subcategory field, displaying options: 'Configure Label', 'Configure Dictionary' (highlighted), 'Configure Styles', 'Show Security Rules', 'Configure Choices', 'Show Choice List', and 'Show - 'u_choice_3''. The browser's address bar shows the URL: dev450742.service-now.com/now/nav/ui/classic/params/target/u_incident_workflow.do%3Fsys_id%3D-1. The top navigation bar includes 'servicenow', 'All', 'Favorites', 'History', 'Workspaces', and a search bar.

- Now scroll down to the choice section.
- Then add the dependent fields for which option dependent on the category gives the category choice backend name to the subcategory choice values.
- Wi-Fi → Network.
- Projector → Hardware. ● Forgot password → Access.

dev450742.service-now.com/now/nav/ui/classic/params/target/sys_dictionary.do%3Fsys_id%3D26d2229d83b3cb10b3fac4b6feaad361%26sysparm_domain%3Dnull%26sysparm_domain_scope...

servicenow All Favorites History Workspaces Dictionary Entry - Subcategory Search

Dictionary Entry Subcategory View: Advanced*

Use dependent field ☒

Dependent on field Category

Create Choice List Delete Column Update

Related Links

Show Table Run Point Scan

Access Controls Choices (4) Attributes (1) Labels (1)

Label Search Actions on selected rows... New

Choices

	Label	Value	Language	Sequence	Inactive	Updated	Dependent value
☐	Forgot Password	forgot password	en	2	false	2026-10-07 01:11:43	
☐	Slow Computer	slow computer	en	3	false	2026-10-07 01:11:43	
☐	Projector	projector	en	1	false	2026-10-07 01:11:43	
☐	Wi-Fi	wi-fi	en	0	false	2026-10-07 01:11:43	

Insert a new row

- Then save the Form.
- Then it will display each choice to each one separately based on the category subcategory displays.

dev450742.service-now.com/now/nav/ui/classic/params/target/u_incident_workflow.do%3Fsys_id%3D-1

servicenow All Favorites History Workspaces Incident Workflow - Create INC0001002 Search

Incident Workflow New record Submit

Number INC0001002

Caller

Category Network

Subcategory -- None --

Short Description -- None --

Description Wi-Fi

State -- None --

Assigned Group

Milestone 3 : Automation using Flow Designer & Email Notification.

When a student creates a new IT support ticket, they only need to enter the Caller Name and Short Description of the issue. The student does not need to manually select the Category or Subcategory. Once the ticket is submitted, a Flow Designer automation is triggered and evaluates the Short Description for predefined keywords such as WiFi, Projector, Password, or Slow. Based on the identified keyword, the flow automatically assigns the appropriate Category and Subcategory to the ticket.

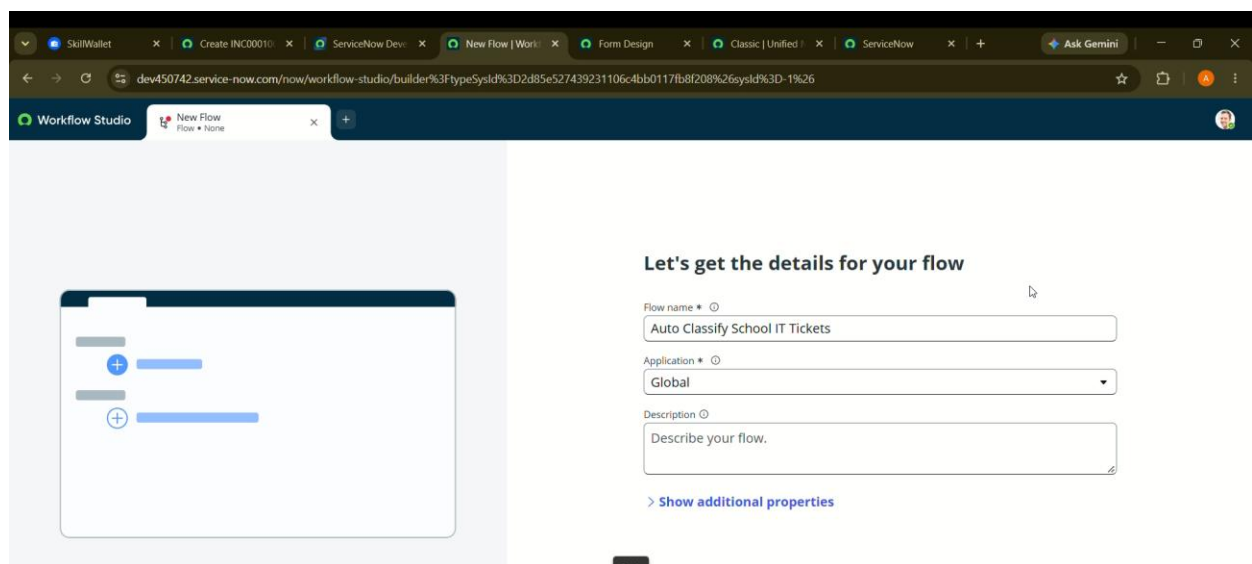
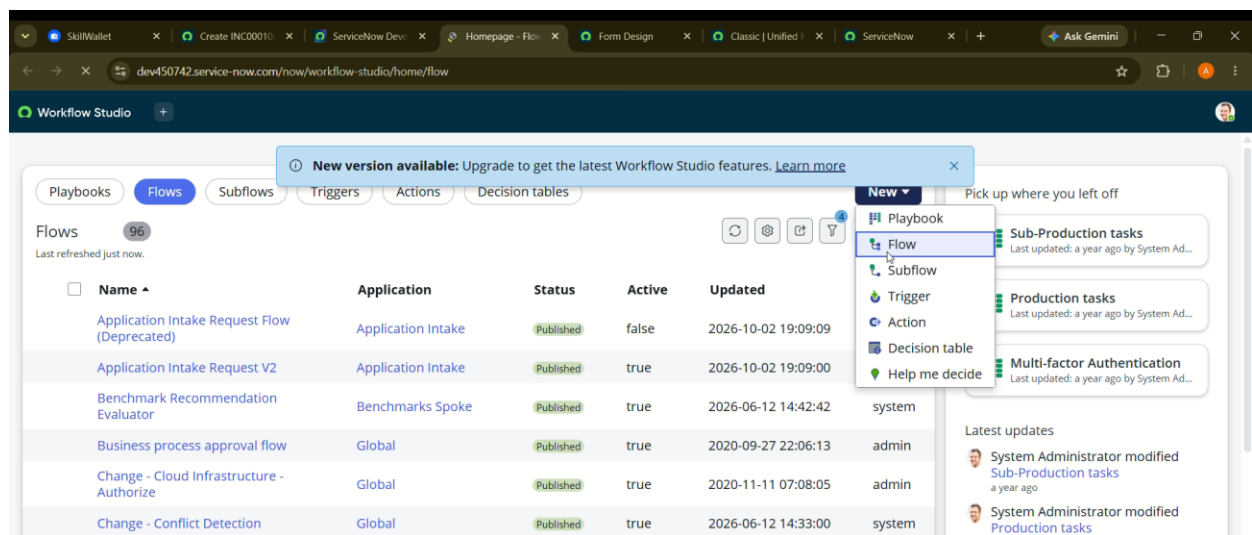
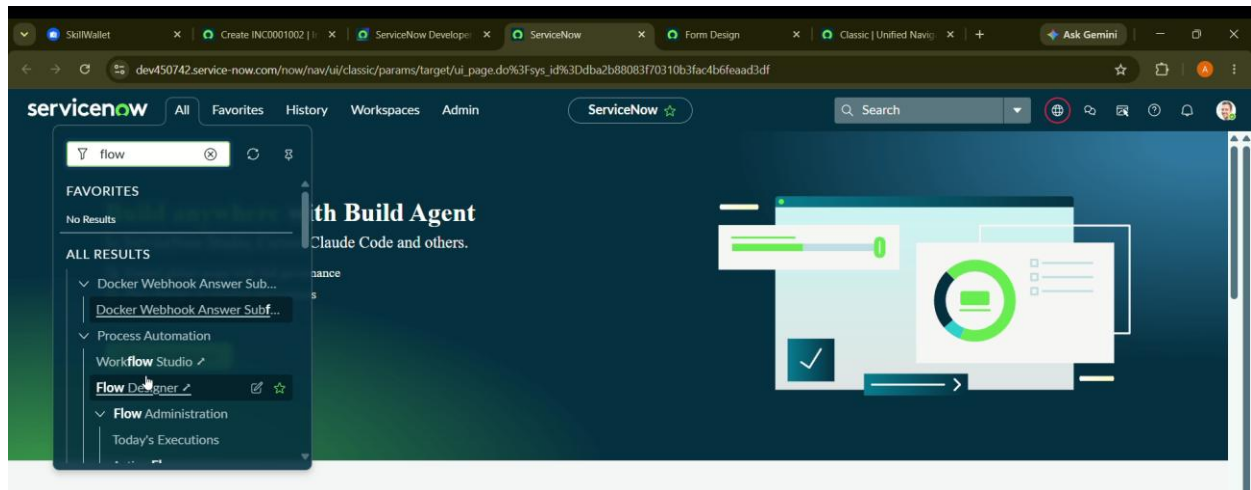
- If the short description contains the word *WiFi*, the flow assigns **Category = Network** and **Subcategory = WiFi**.
- If the description contains *Projector*, the flow assigns **Category = Hardware** and **Subcategory = Projector**.
- If the description includes *Password* or *Login*, the flow assigns **Category = Access** and **Subcategory = Forgot Password**.
- If the description mentions *Slow* or *Hanging*, the flow assigns **Category = Performance** and **Subcategory = Slow Computer**.

This automation ensures consistent classification, eliminates human error, and significantly reduces the manual effort required by IT staff.

The entire process is implemented using **Flow Designer**, making the solution easy to maintain, scalable, and completely no-code.

Activity : 1 Auto Classification using Flow design.

- Go to **All menus** → Search **Flow designer**.
- Under **Process Automation** Open the **Flow designer**.
- On the top right corner Click **New** and Select **Flow** Option.
- Name : **Auto Classify School IT Tickets**.
- Application : **Global**.
- Now Click the **Build Flow** button.



Activity : 2 Trigger Configuration.

- Trigger : **Record Created.**
- Table : **Incident Workflow.**
- Condition : **Category is Empty.**
- Then Click **Done.**

The screenshot shows the ServiceNow Workflow Studio interface. The browser address bar displays the URL: `dev450742.service-now.com/now/workflow-studio/builder%3Ftable%3Dsys_hub_flow%26sysid%3D0616ae1583f3cb10b3fac4b6fead39e%26`. The workflow is titled "Auto Classify School IT Tickets" and is currently "Inactive".

TRIGGER

Incident WorkFlow Created where (Category is)

Trigger: Created

* Table: Incident WorkFlow [u_incident_w...]

Condition: Category is NULL

Buttons: Delete, Cancel, Done

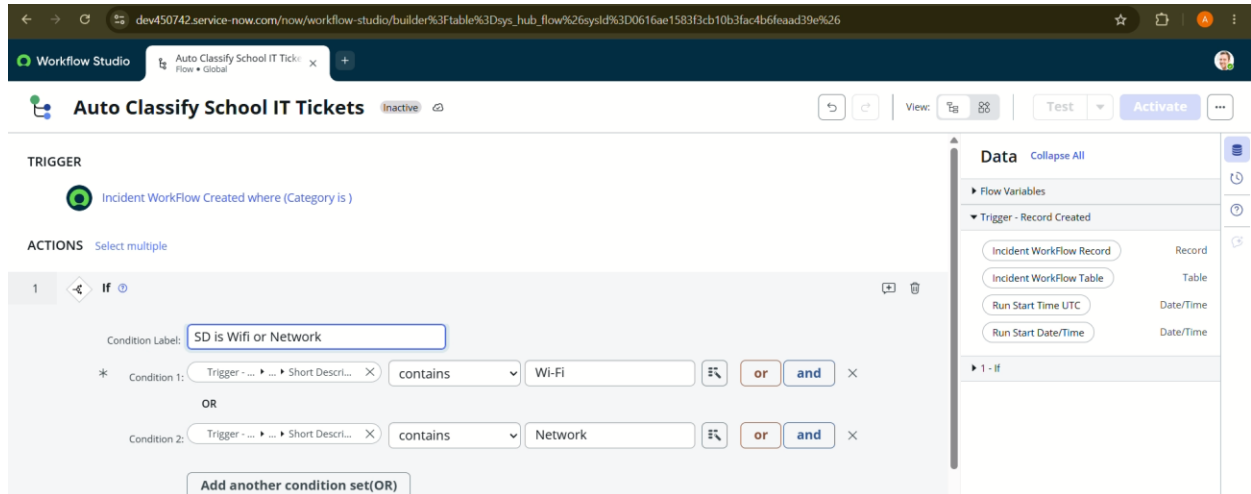
Advanced Options

- Now Under **Actions** Select **Flow Logic** Option.

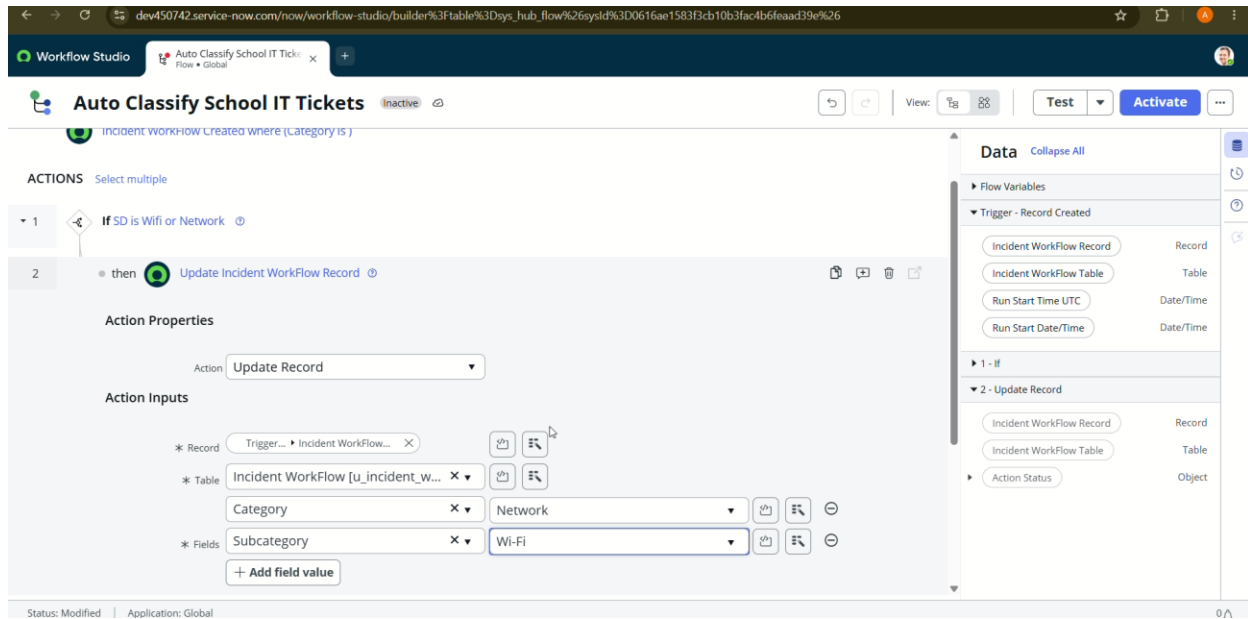
Activity : 3 Wi-Fi Issue Classification.

- Select **If** condition.
- Condition label : SD is **Wifi or Network** (Give as ur wish).
- Condition 1 : Using data pill picker Select **Short description** as Field , and operator **Contains** , Value is **Wi-Fi**.
- Condition 2 : Using data pill picker Select **Short description** as Field, and Operator is **Contains**, Value is **Network**.

- Then click **Done** Button

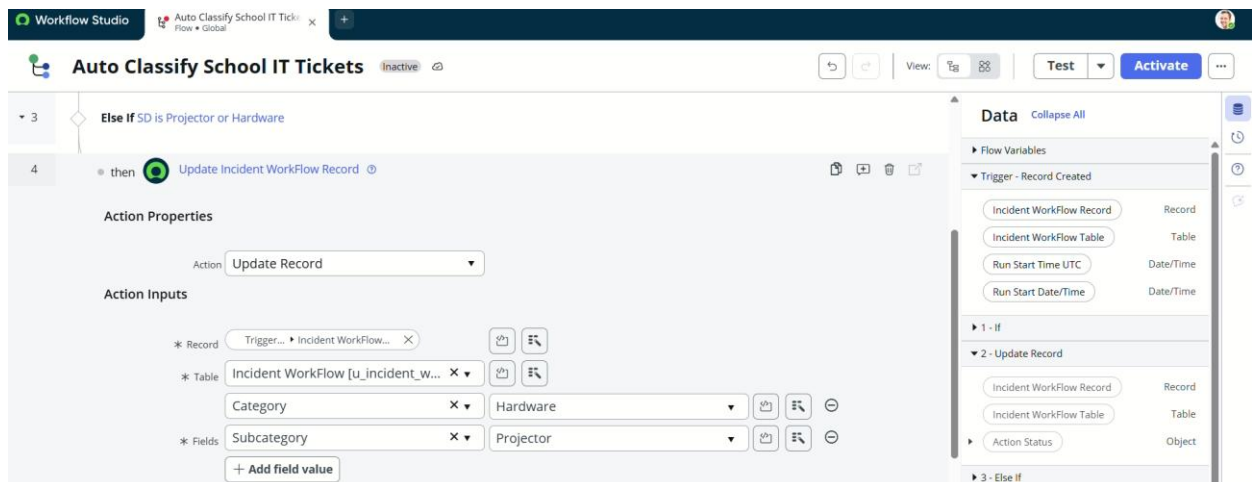


- Now Click **Action** to **Update record**.
- Action Inputs.
- In Record Field drag and drop data from the right side.
- Table : Auto updates as Incident Workflow.
- Fields Set **Category = Network**.
- Set **Subcategory = Wi-Fi**.
- Then Click the **Done** button.



Activity : 4 Projector Issue Classification.

- Now Select the **Else If** Condition.
- Condition label : SD is **Projector or Hardware**.
- Condition : **Short description Contains Projector**.
- Then Click the **Done** button.
- Now Action is an **Update record**.
- Set **Category = Hardware**



- Then Click the **Done** button.

Activity : 5 Password Issue Classification.

- Now again select **Else If** condition.
- Condition label : SD is **Forgot password**.
- Condition : **Short description contains Forgot password**.
- Then Click the **Done** button.
- Now Action is an **Update Record**.
- Set **Category** = **Access**.
- Set **Subcategory** = **Forgot Password**.
- Then Click the **Done** button.

The screenshot displays the ServiceNow Workflow Studio interface for a workflow named "Auto Classify School IT Tickets". The workflow is currently in an "Inactive" state. The main canvas shows a sequence of steps:

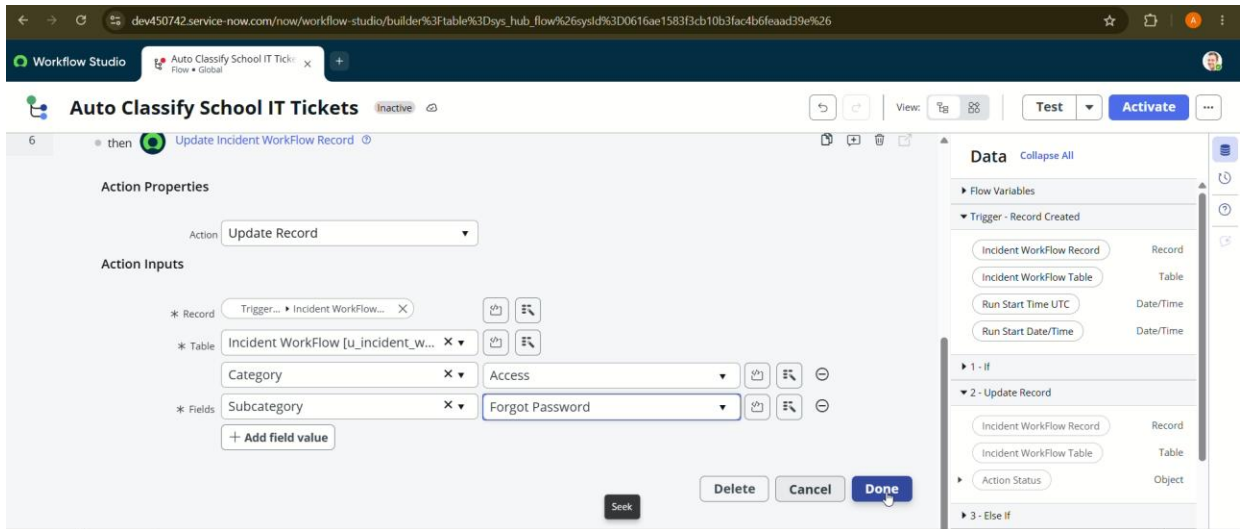
- Step 3: A diamond-shaped decision node labeled "Else If SD is Projector or Hardware".
- Step 4: A "then" block containing an "Update Incident WorkFlow Record" action.
- Step 5: A new "Else If" condition node being configured.

The configuration for Step 5 is as follows:

- Condition Label:** SD is Forgot password
- Condition 1:** Trigger - ... -> Short Descri... (with a dropdown menu set to "contains" and a text input field containing "Forgot Password").
- Logic:** The conditions are connected by "or" and "and" operators.
- Buttons:** "Add another condition set(OR)", "Seek", "Delete", "Cancel", and "Done".

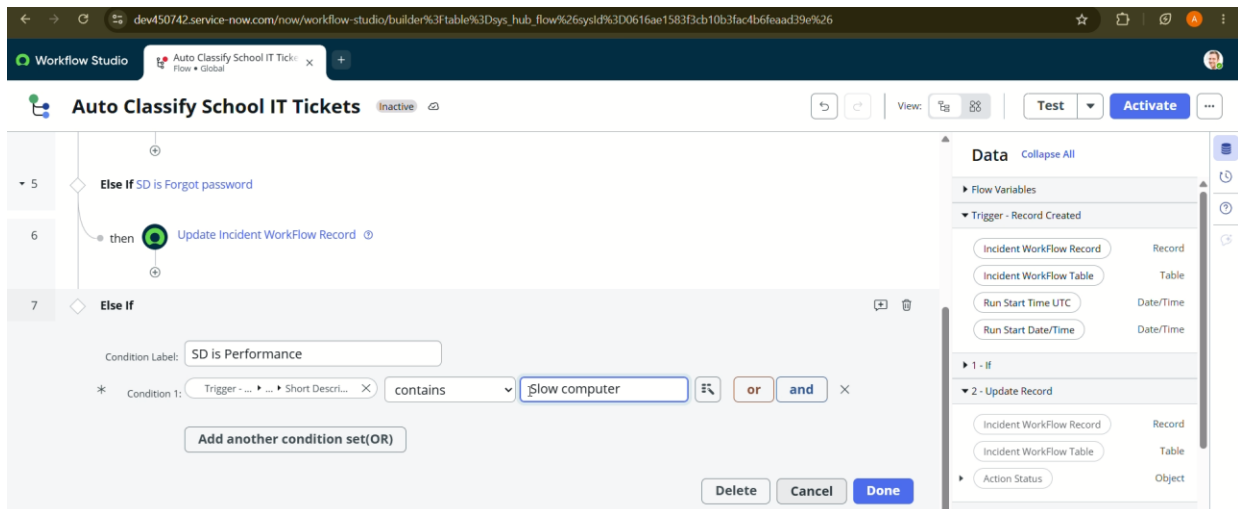
On the right side, the "Data" panel is visible, showing a list of variables and their types:

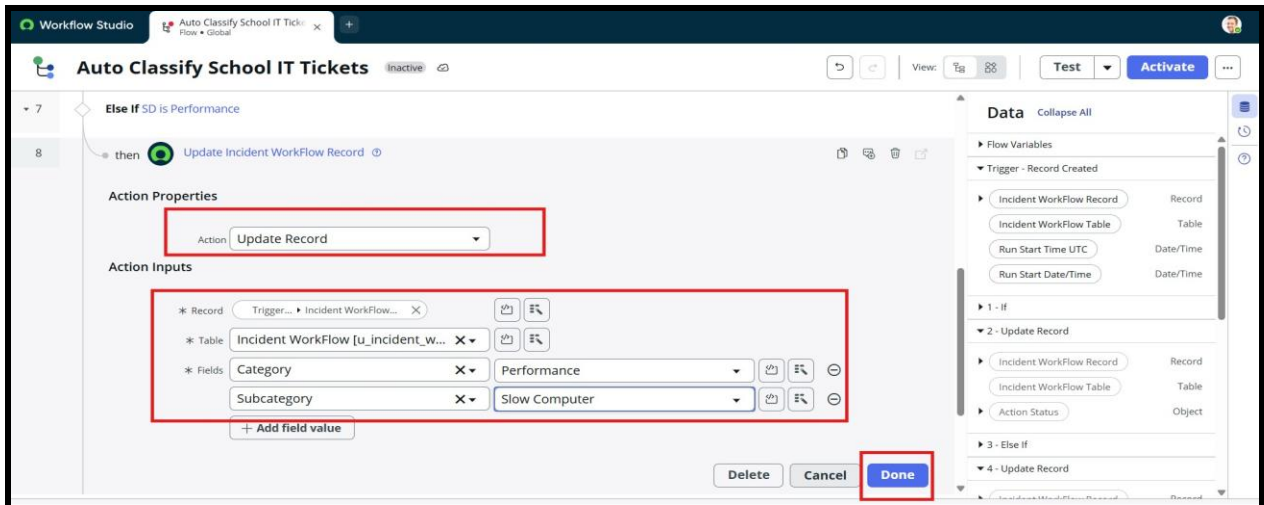
- Flow Variables:**
 - Incident WorkFlow Record (Record)
 - Incident WorkFlow Table (Table)
 - Run Start Time UTC (DateTime)
 - Run Start Date/Time (DateTime)
- 1 - If:**
 - Incident WorkFlow Record (Record)
 - Incident WorkFlow Table (Table)
- 2 - Update Record:**
 - Incident WorkFlow Record (Record)
 - Incident WorkFlow Table (Table)
- Action Status:** Object
- 3 - Else If:**



Activity : 6 Slow Computer Classification.

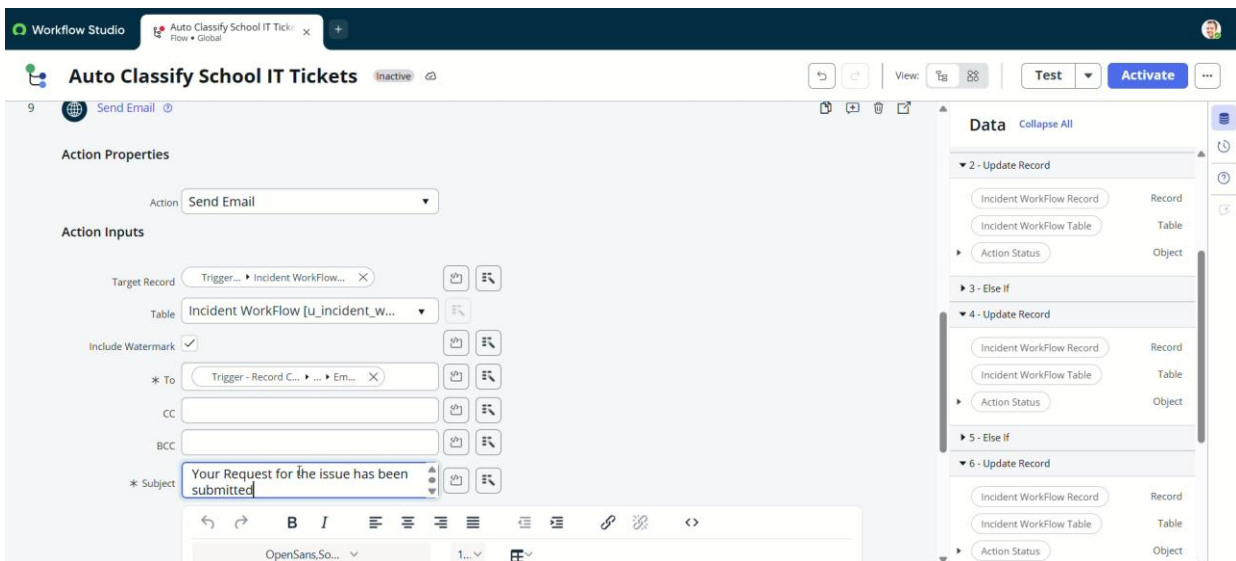
- Now Again Click the **Else If** Condition.
- Condition label : **SD is Performance**.
- Condition : **Short description contains Slow Computer**.
- Then Click the **Done** button.
- Now Action is an **Update Record**.
- Set **Category = Performance**.
- Set **Subcategory = Slow computer**. • Then Click the **Done** button.

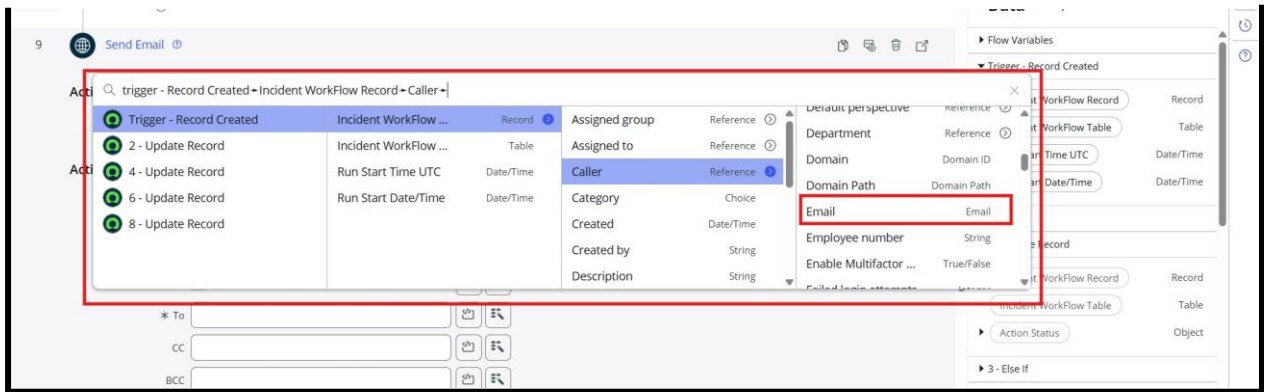




Activity : 7 Email Action Setup.

- After classification, an email notification is sent using Flow Designer.
- Set Action **Send Email**.
- To : **Caller** → **Email**.
- Subject : **Your Request for the issue has been submitted.**
- Body : **Ticket confirmation message.**
- Then Click the **Done** button.





- Now Click the **Activate** button from the top right corner.

PHASE 6 : Testing, Validation & Security.

- The objective of Phase 4 is to ensure that the developed solution works as expected, meets all functional requirements, and follows basic security and access control best practices.
- This phase validates the end-to-end automation, confirms data accuracy, and ensures that only authorized users can access or modify ticket data.

Data Validation Checks

- Mandatory fields captured correctly.
- Auto-number generated without duplication.
- Category and Subcategory values stored accurately.
- Reference fields resolve correctly to user and group tables.

Activity : 1 Testing & Validation.

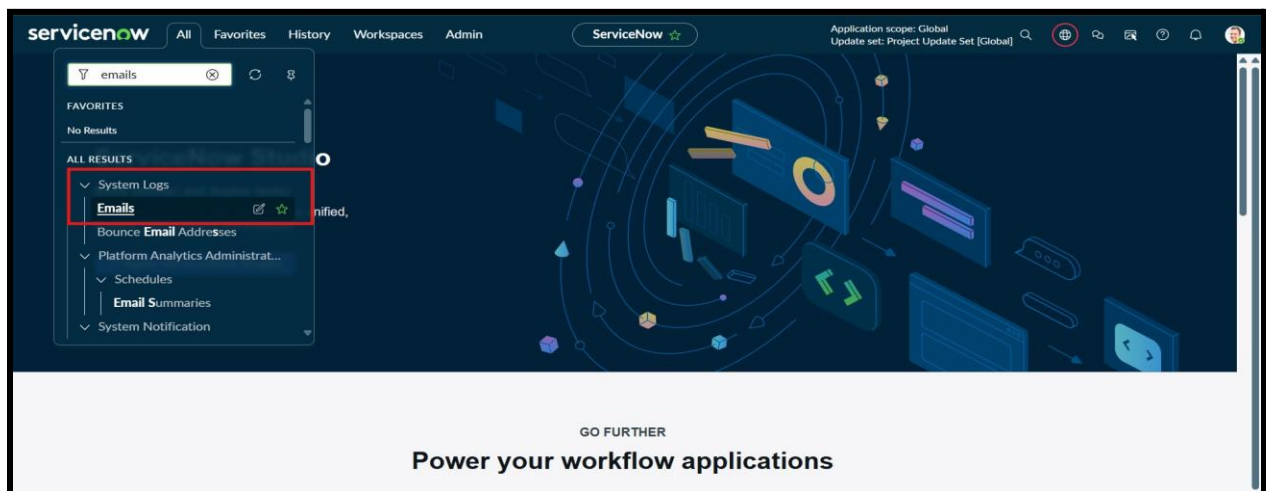
Test Scenario 1

- Enter one Caller Name.
- Short Description: WiFi not working in library • Then Save the Form.
- And Reload the Form.

- Then it fills auto as :
- Category → **Network.**
- Subcategory → **WiFi.**

Activity : 2 Check Email Notification.

- Open All menus → Search Emails.
- Under System logs → Select Emails.
- Search with the Given Subject Name.
- Open that Record and Under Related Links Click Preview mail Option.



Created	Recipients	Subject	Type	Notification type	User ID
2026-02-06 23:04:59	savannah.kesich@example.com	Your Request for the issue has been submitted.	send-ready	SMTP	(empty)
2026-02-06 23:04:52	eliseo.wice@example.com	Your Request for the issue has been submitted.	send-ready	SMTP	(empty)

Preview Email

Hi greetings from xyz college!!!

We are viewed your request and solved by short time period.,
Thankyou ,

Ref:MSG0000396_ChBc6sAUD3zwGS0qlNyn

Message-ID: <983449560443704478999008@super120160.wor101-service-now.com>
X-ServiceNow-Source: FlowDesigner-dad2c02bf3763210749db0a6a8f4e75c
X-ServiceNow-Generated: true
X-ServiceNow-SysEmail-Version: 2

Related Links
[Preview Email](#)

Activity : 3 Test Scenario 2

- Enter one Caller Name.
- Short Description: Projector not turning on.
- Then Save the Form.
- And Reload the Form.
- Then it fills auto as :
- Category → **Hardware.**
- Subcategory → **Projector.**
- Then Email sent to caller Mail id.

Incident Workflow
INC00512

Number: INC00512

* Caller: Sam Sorokin

Category: Hardware

Subcategory: Projector

Short description: Projector not turning on

Description:

State: -- None --

Assigned group:

Assigned to:

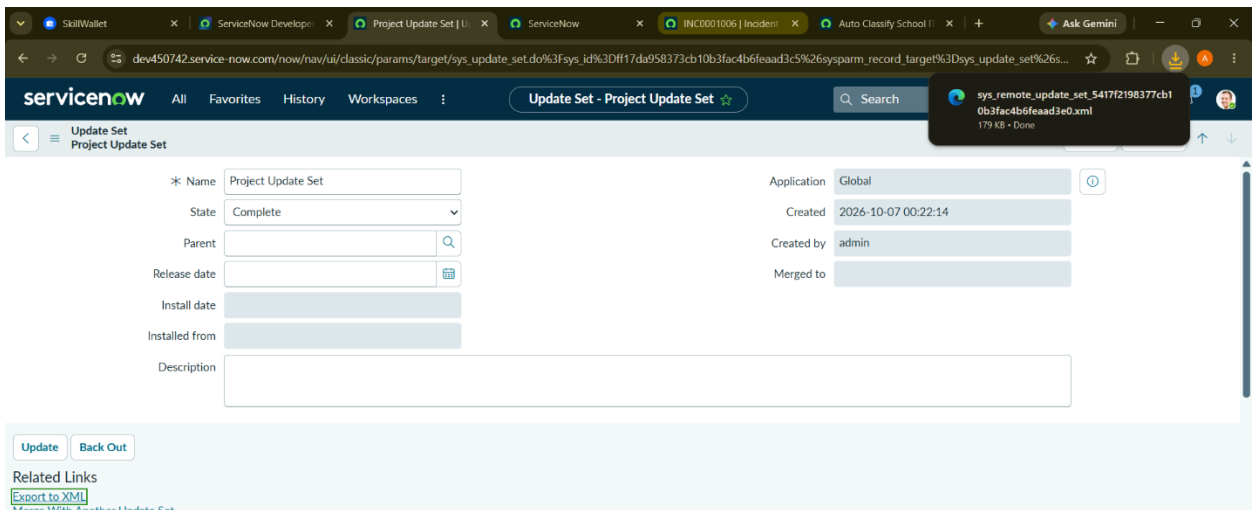
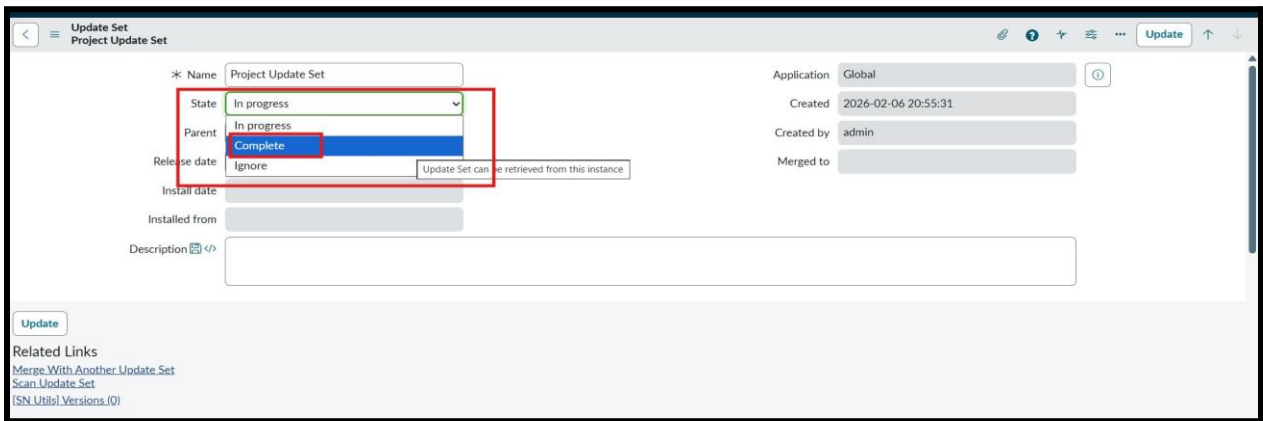
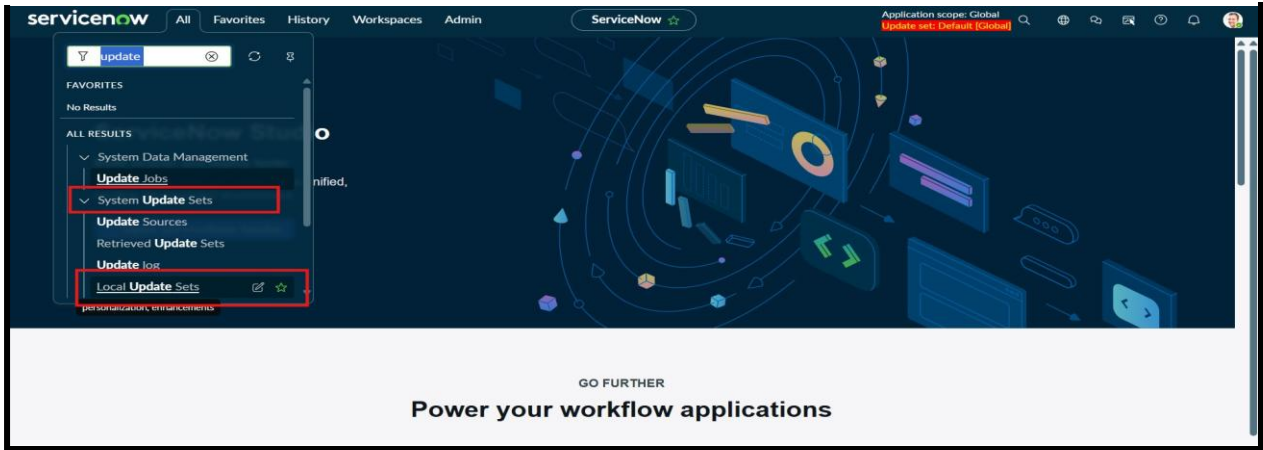
Update Delete

PHASE 5: Deployment & Final Documentation.

- The objective of Phase 4 is to deploy the developed solution to the target environment and finalize all project documentation.
- This phase ensures that the solution is stable, reusable, and ready for demonstration or production use, along with complete and clear documentation for future reference.

Activity : 1 Make Update set to Complete State.

- Go to All menus → Search For **Local Update Sets**.
- Under **System Update sets** Select → **Local Update sets**.
- Open the **Project Update set**.
- Make the State From **In progress** to **Complete**.
- Then **Save** the Form.
- Under Related links Click **Export to XML** Option.
- Then the Xml Format the Update set data will be **downloaded** to your system.
- We can share through WhatsApp or Mail that downloaded file.
- Then the Created Update set will transform to others in this way.



Update Set
Project Update Set

* Name: Project Update Set

State: Complete

Parent:

Release date:

Install date:

Installed from:

Description:

Application: Global

Created: 2026-02-06 20:55:31

Created by: admin

Merged to:

Update Back Out

Related Links

[Export to XML](#)

[Generate With Abstraction Update Set](#)

[Scan Update Set](#)

[\(SN Utils\) Versions \(0\)](#)

Conclusion

- The **Auto Ticket Classification using Flow Designer** project successfully delivers a complete, end-to-end automation solution for a school IT helpdesk. The primary objective of reducing manual effort in ticket categorization and improving response efficiency has been fully achieved.
- Through structured requirement analysis, careful backend configuration, and no-code automation using Flow Designer, the system now automatically classifies IT tickets based on the short description provided by students. The implementation of dependent choice fields ensures data accuracy, while automated email notifications improve communication by instantly acknowledging ticket creation to the caller.
- The solution was developed using a phased approach, ensuring clarity, stability, and ease of maintenance. Comprehensive testing and security validation confirmed that the system functions reliably, maintains data integrity, and enforces appropriate access controls.
- Overall, this project demonstrates how ServiceNow can be effectively used to build scalable, maintainable, and user-friendly automation solutions without relying on complex scripting or machine learning. The design is well-suited for small to medium environments such as educational institutions and can be easily extended in the future to include advanced features like assignment automation, SLA tracking, or predictive intelligence.